Earth vs Mustafar

STAC Wars report. Team: Epstein’s list. By Yamini and Yashika.

Introduction

For this project we built a two-stage rocket in OpenRocket and flew it in two places: Earth's normal atmosphere and Mustafar. Both simulations are saved in our .ork file, called "earth baseline" and "mustafar". The rocket is exactly the same in both. It has the same parts, motors, parachutes, launch rod and 10 m/s wind. Only the environment is different, so any change in the flight is because of that.

For each thing we changed, we wrote down what we expected first and then checked it against the simulation. We also did a few quick hand calculations to make sure the simulation numbers made sense. All the results are for the sustainer (the top stage) unless we say booster.

Results

Table 2 puts the Earth run next to the full Mustafar run (everything changed together).

Sustainer	Earth	Mustafar
Apogee	1197 m	4078 m
Time to apogee	15.0 s	39.9 s
Max velocity	185.5 m/s	199.6 m/s
Max Mach number	0.55	0.38
Stability at rod exit	1.77 cal	1.77 cal
Stability during climb	1.98 to 3.15 cal	1.97 to 3.12 cal
Descent speed, drogue	22.6 m/s	32.5 m/s
Descent speed, main chute	did not open	9.6 m/s
Ground hit speed	24.7 m/s	13.9 m/s
Landing distance from pad	117 m	812 m
Flight time	68.5 s	315.9 s
Booster descent speed	1.84 m/s	2.72 m/s
Booster landing distance	5960 m	7040 m

Table 2. Earth baseline vs Mustafar.

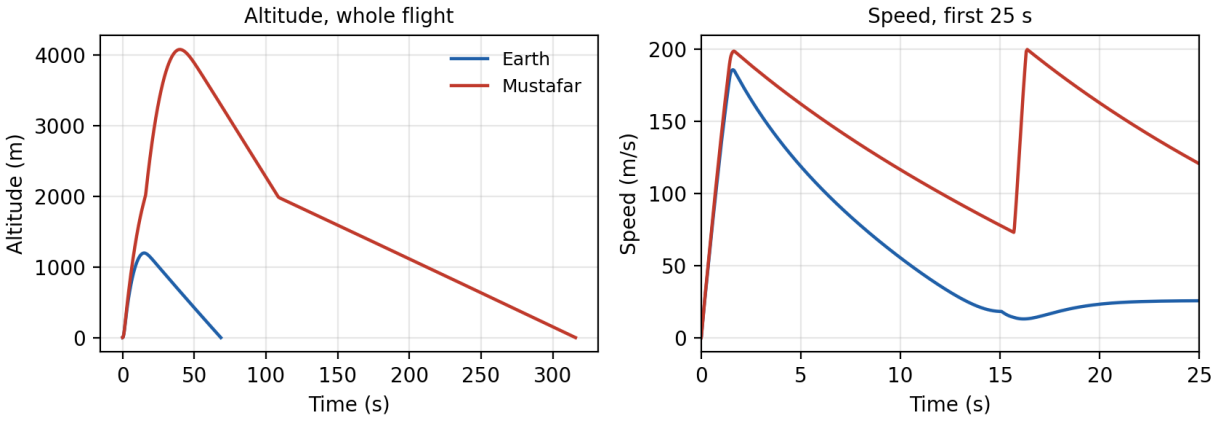


Figure 1. Altitude and speed on Earth and Mustafar.

Testing each change:

Gravity (9.81 to 6.5 m/s^2)

What we expected. The rocket weighs about 66% of what it does on Earth. The motors give the same thrust, so it should lose less speed to gravity after burnout, coast higher and take longer to reach apogee. A parachute should also bring it down a bit slower.

What happened. Time to apogee went from 15.0 s to 39.9 s. Gravity didn't change the top speed much, though. The burn is only 1.69 s in both runs, so gravity only saves about (9.79 − 6.5) × 1.69 = 5.6 m/s. The real gain was about 14 m/s, so the rest came from less drag. The big gravity effect is in the coast. With no drag, a rocket going 190 m/s would coast 1840 m on Earth and 2780 m on Mustafar, which is about 1.5 times more (9.81 / 6.5 = 1.5).

Thinner air (lower pressure and density)

What we expected. Mustafar air is only about 30% as dense as Earth air, so drag should be about 70% lower at the same speed. The rocket should go higher and faster. But parachutes lose drag too, so we expected them to fall faster, by about 1.48 times (from √(6.5 × 1.225 / (9.81 × 0.373))). We thought the stability margin wouldn't change because it depends on the shape of the rocket and not the air, but the force that keeps the rocket straight should be a lot weaker.

What happened. The drogue descent was 22.6 m/s on Earth and 32.5 m/s on Mustafar, a ratio of 1.44. We also did it by hand using $v = \sqrt{(2mg / (\rho C_d A))}$ with m = 6.2 kg, C_d = 1.55 and a 0.4 m chute, and got 22.6 and 33.3 m/s, so the simulation agrees with the physics. The booster chute went from 1.84 to 2.72 m/s (1.48 times). Stability at rod exit was 1.77 calibers in both runs and stayed between about 2.0 and 3.1 during the climb in both. The dynamic pressure at top speed is about 21 kPa on Earth and 7.4 kPa on Mustafar (ignoring how density changes with height), so roughly a third. That means the fins have much less force to work with even though the stability number is the same.

Temperature (288 K to 700 K)

What we expected. The speed of sound is √(γRT). That is about 340 m/s at 15 °C and about 530 m/s at 700 K. So the same speed should give a lower Mach number on Mustafar. Hotter air at the same pressure is also less dense, which adds to the thin-air effect.

What happened. Top speed went up about 8% but max Mach dropped from 0.55 to 0.38. By hand, 199.6 / 530 = 0.38, so that matches. We never get close to Mach 0.8, where drag shoots up, so the lower Mach number doesn't change the drag much for our rocket. Mostly the temperature matters through density. One thing OpenRocket doesn't do is heat. At about 427 °C a cardboard body and nylon parachutes wouldn't survive in real life.

Wind (10 m/s side wind)

What we expected. The rocket leaves the rod at about 18 m/s into a 10 m/s wind, so the angle of attack is about atan(10/18) = 29°. After that it should turn into the wind (weathercock) and lose a little altitude. Because the air is thinner on Mustafar we thought it would turn back more slowly. On the parachute the rocket moves with the wind, so drift should be about wind speed × time in the air. Mustafar's descent is much longer, so it should land further away.

What happened. Right after the rod the angle of attack was about 24° on both worlds and it dropped to around 1° within about a second. At burnout the rocket was tilted 11.7° from vertical on Mustafar and 13.5° on Earth, so the thin air did not make it correct more slowly. We think the rocket is going so fast by then that it doesn't matter. The drift was much clearer. On Mustafar the rocket was 1689 m upwind at apogee and landed 812 m downwind, so it travelled about 2500 m in 276 s (about 9.1 m/s). On Earth it travelled about 459 m in 53.5 s (8.6 m/s). So in both cases it drifts at about 90% of the wind speed and the

only difference is how long it is in the air. The booster did the same thing: 5960 m on Earth and 7040 m on Mustafar.

Conclusion

Most of our predictions held. Lower gravity gave a longer climb. Thinner, hotter air cut the drag, so the rocket went faster and higher, and the parachutes fell about 40 to 50% faster than on Earth. The higher temperature raised the speed of sound, so the Mach number dropped from 0.55 to 0.38 even though the rocket was going faster.

The prediction that did not clearly hold was weathercocking. Thin air did not make the rocket correct more slowly, probably because it was already moving fast.